Chapter 1 Rings

1.1 Definitions

1.2 UFD,PID,Euclidean Ring

1.3 Localization

1.4 title

1.4.1 Ring Maps of Finite Type and of Finite Representation

Definition 1.4.1.1 Let $R \to S$ be a ring map:

- $R \to S$ is **of finite type**, or that S is a **finite type R-algebra** if there exist an $n \in \mathbb{N}$ and an surjection of R-algebras

$$R[x_1, \dots, x_n] \to S$$

- $R \to S$ is **of finite presentation** if there exist integers $n, m \in \mathbb{N}$ and polynomials

$$f_1, \dots, f_m \in R[x_1, \dots, x_n]$$

and an isomorphism of R-algebras

$$R[x_1, \dots, x_n]/(f_1, \dots, f_m) \cong S$$

1.4.2 Ring Maps of Finite Type and of Finite Presentation

1.4.3 Finite Ring Maps

1.4.4 Homomorphisms Essentially of Finite Type

Definition 1.4.4.1 Let $R \to S$ be a ring map:

- $R \to S$ is **essentially of finite type** if S is the localization of an R-algebra of finite type. i.e.

$$S = M^{-1}R[x_1, \dots, x_n]$$

- $R \to S$ is **essentially of finite representation** if S is the localization of an R-algebra of finite representation

◇ **Lemma 1.4.4.2** The class of ring maps which are **essentially of finite type** is preserved under composition. Similarly for essentially finite representation.

∘ **Proof** ... □

Under Base CHange

♠ **Proposition 1.4.4.3** ...

∘ **Proof** ... □

Maps to Artinian Local Ring

◇ **Lemma 1.4.4.4** ...

∘ **Proof** ... □

Essentially Finite Maps between Local Rings

◇ **Lemma 1.4.4.5** If $\varphi : R \to S$ be essentially of finite type with R and S local(but, φ may be not local) . Then, there exists an n and a maximal ideal

$$m \subseteq R[x_1, \dots, x_n]$$

lying over m_R such that: S is a localization of a quotient $R[x_1, \dots, x_n]_m$.

∘ **Proof** ... □

1.5 Nilpotent, Idempotent and

1.5.1 Nilpotent Elements and Nilradicals

1.5.2 Jacobson Radical

1.6 Change of Rings

Definition 1.6.0.1 ...

Chapter 2 Modules

2.1 Internal Hom

2.2 Tensor Product

Definition 2.2.0.1

◇ **Lemma 2.2.0.2** (Tensor Product commutes with Colimits) Let (M_i, μ_{ij}) be a system over a preordered set I, then

$$colim(M_i \otimes N) \cong (ColimM_i) \otimes N$$

∘ **Proof** □

◇ **Lemma 2.2.0.3** Let $M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$ be an exact sequence of R-modules and homomorphisms, and N be an R-module, then the sequence

$$M_1 \otimes N \xrightarrow{f\otimes id} M_2 \otimes N \xrightarrow{g\otimes id} M_3 \otimes N \longrightarrow 0$$

is exact. In other word, $- \otimes N$ is right exact.

∘ **Proof** □

2.2.1 Finiteness

◇ **Lemma 2.2.1.1**

∘ **Proof** □

2.2.2 Tensor Algebra

2.3 Limits, Colimits

2.3.1 Directed System, Directed Limits

Definition 2.3.1.1 Let $(I, \leq)$ be preordered set. Then, (M_i, μ_{ij}) of R-modules over I consists of a family of R-modules $\{M_i\}$ indexed by I, and a family of R-module maps $\{\mu_{ij} : M_i \to M_j\}_{i\leq j}$ such that for $i \leq j \leq k$

$$\mu_{ii} = id_{M_i} \qquad \mu_{ik} = \mu_{jk} \circ \mu_{ij}$$

We say (M_i, μ_{ij}) is a **directed system** if I is a directed set.

◇ **Lemma 2.3.1.2** Let (M_i, μ_{ij}) be a family of R-modules over the preordered set I. The colimit o the system (M_i, μ_{ij}) is the quotient R-module

$$(\oplus_i M_i)/Q \qquad Q =< \{\iota_i(x_i) - \iota_j(\mu_{ij}(x_i))\}_{i \leq j} >_R$$

where $\iota_i : M_i \to \oplus_{i \in I} M_i$ is a natural inclusion.
We denote the colimit

$$colimit(M) = colim_i M_i$$

Denote $\pi : \oplus_{i \in I} M_i \to M$ the projection map and $\phi_i = \pi \circ \iota_i : M_i \to M$.

∘ **Proof** ... □

◇ **Lemma 2.3.1.3** Let (M_i, μ_{ij}) be a system of R-modules over the preordered set I.
Assume that I is directed. The colimit of the system $(M_i.\mu_{ij})$ is canonically isomorphic to the module M defined as follows:

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∘ **Proof** ... □

◇ **Lemma 2.3.1.4** Let (M_i, μ_{ij}) be a directed system. Let $M = colim M_i$ with

$$\mu_i : M_i \to M$$

Then, $\mu_i(x_i) = 0, x_i \in M \Leftrightarrow$ There exists $j \geq i$ such that $\mu_{ij}(x_i) = 0$

∘ **Proof** ... □

Morphisms of Directed Systems

◇ **Lemma 2.3.1.5** ...

∘ **Proof** ... □

◇ **Lemma 2.3.1.6** ...

∘ **Proof** ... □

2.3.2 Directed Limit in Index Category

♠ **Proposition 2.3.2.1** ...

∘ **Proof** ... □

2.4 Characterizing Finite and Finitely Presented Modules

2.5 Finite Case of Modules and Maps

2.5.1 Finite Modules, and Finitely presented modules

Definition 2.5.1.1 Let R be a ring and M is R-module:

- We say M is a **finite** R-module, or finitely generated module if there exists $n \in \mathbb{N}$ and $x_i \in M$, with exact sequence

$$\oplus Rx_i \to M \to 0 \quad \text{or} \quad R^{\oplus n} \to M \to 0$$

- M is **finitely represented** if there exists $m, n \in \mathbb{N}$ and exact sequence

$$R^{\oplus m} \to R^{\oplus n} \to M \to 0$$

  which implies:

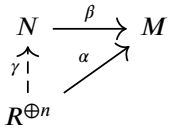
  - M is finitely generated;
  - The relation between the generators are also finitely generated.

  such choice of exact sequence is called a **presentation**.

◇ **Lemma 2.5.1.2** Let R be a ring, and

$$\alpha : R^{\oplus n} \to M \qquad \beta : N \to M$$

be R-module maps.If $Im(\alpha) \subseteq Im(\beta)$, then there exists an R-module map $\gamma : R^{\oplus n} \to N$ such that $\alpha = \beta \circ \gamma$.



◦ **Proof** Let $\alpha(e_i) = \beta(n_i)$. Then, consider the mapping

$$\gamma : R^{\oplus n} \to N \qquad (a_1, \dots, a_n) \mapsto \sum a_i n_i$$

□

◇ **Lemma 2.5.1.3** Let R be a ring, and $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of R-modules:

- If M_1 and M_3 are finite R-modules, then M_2 is a finite R-module;
- If M_1 and M_3 are finitely presented R-modules, then M_2 is finitely presented;
- If M_2 is a finite R-module, then M_3 is a finite R-module;
- If M_2 is finitely presented R-module and M_1 is a finite R-module, then M_3 is a finitely represented R-module;
- If M_3 is finitely presented R-module and M_2 is a finite R-module,then M_1 is a finite R-module.

◦ **Proof**

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□

◇ **Lemma 2.5.1.4** Let R be a ring and let M be a finite R-module. There exists a filtration by finite R-module

$$0 \subseteq M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

such that each quotient M_i/M_{i-1} is isomorphic to R/I_i for some ideal I_i of R.

∘ **Proof** ... □

◇ **Lemma 2.5.1.5** Let $R \to S$ be a map, and M be an S-module. If M is finite as an R-module, then M is finite as an S-module.

∘ **Proof** ... □

2.6 Base Change

Chapter 3 Associative Algebra

Chapter 4 Polynomial Algebra

4.1 Discriminant

Chapter 5 Matrix Algebra

The next part of contents is Noncommutative ring theory.
Reference

- T.Y.Lam
- Noncommutative Noetherian Ring, J.C.McConnell

Chapter 6 Free Modules, Projective and Injective Modules

6.1 Free Modules

6.1.1 IBN

Category of Left/Right Modules

Definition 6.1.1.1

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Free Module

◇ **Lemma 6.1.1.2** (Generation Lemma)

∘ **Proof** ... □

★ **Corollary 6.1.1.3**

∘ **Proof** ... □

IBN Properties

Definition 6.1.1.4 ...

6.1.2 Stable Finiteness

Definition 6.1.2.1 Recall: A ring is **Dedekind-finite** if

$$c, d \in S, cd = 1 \Rightarrow dc = 1$$

A ring R is **stably finite** if the matrix $M_n(R)$ are Dedekind-finite.

Remark Dedekind finiteness: In $M_n(F)$, we have

$$AB = I \Rightarrow BA = I$$

With:

...

But for $M_\omega(F)$, a counterexample is

...

♠ **Proposition 6.1.2.2** The following properties of R are equivalent:

- R is stably finite;
- For any n, $R^n \cong R^n \oplus N \Rightarrow 0$ in $\mathfrak{m}_R$;
- For any n, any epimorphism $R^n \to R^n$ in $\mathfrak{m}_R$ is an isomorphism.

∘ **Proof** ... □

♠ **Proposition 6.1.2.3** For any nonzero element R, stable finiteness implies IBN. But not conversely.

∘ **Proof** ... □

♠ **Proposition 6.1.2.4** ...

∘ **Proof** ... □

6.1.3 The Rank Condition

Definition 6.1.3.1

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♠ **Proposition 6.1.3.2** ...

∘ **Proof** ... □

6.1.4 The Strong Rank Condition

6.2 Projective Modules

6.2.1 Definitions, Examples

Definition 6.2.1.1 ...

6.3 Injective Modules

Chapter 7 Flat Modules and Homological Dimension

7.1 Flat and Faithfully Flat Modules

Definition 7.1.0.1 ...

Chapter 8 Jacobson Radical

Definition 8.0.0.1 (Jacobson Radical) Define: The Jacobson radical

$$rad(R) = \cap_{I \in L} I \qquad L = \{I \subseteq R | I \text{ is the left ideal of } R\}$$

◇ **Lemma 8.0.0.2** For $y \in R$, the following statements are equivalent:

- $y \in rad(R)$;
- $1 - xy$ is left-invertible for any $x \in R$;
- $yM = 0$ for any simple left R-module M.

∘ **Proof**

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□

Annihilators

Definition 8.0.0.3 ...

Jacobson Semisimple

Nilpotent

8.1 Jacobson Radical Under Change of Rings

8.2 Group Ring and the J-Semisimplicity

Chapter 9 Division Algebra

Reference

- A First Course in Noncommutative Rings, T.Y.Lam,Chapter 5;
-

9.1 Finite Dimensional Division Algebra

9.1.1 Discussion of Complex Numbers,Quaternion and Frobenius's Theorem

Introduction We have the extension of field

$$\mathbb{R} \rightsquigarrow \mathbb{C} \qquad dim_{\mathbb{R}}(\mathbb{C}) = 2$$

Morever, we want to find some $\mathbb{R}$-algebras, and be able to devide nubmers.

Definition 9.1.1.1 A unital algebra D in which every nonzero element is invertible is called a division algebra.
i.e. for every pair $a \in D - \{0\}$, such x, y exists.

$$ax = 1 \qquad ya = 1$$

A question is:

Is it possible to define multiplication on an n-dimensional real space so that it becomes a real division algebra?

Standard Results:

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Assume: D is an n-dimensional $\mathbb{R}$-algebra.

◇ **Lemma 9.1.1.2** For every $x \in D$, there exists a $\beta \in \mathbb{R}$ such that

$$x^2 + \beta x \in \mathbb{R}$$

∘ **Proof** ... □

Define:

$$V := \{v \in D | v^2 \leq 0\}$$

◇ **Lemma 9.1.1.3** If $u, v \in V$ are linearly independent, then so are u, v and 1.

∘ **Proof** ... □

Jordan Product

Definition 9.1.1.4 (Jordan Product) For $x, y \in D$, we write

$$x \circ y := xy + yx$$

Frobenius's Theorem

▶ ***Theorem*** (Frobenius's Theorem) A **finite dimensional division algebra** D is isomorphic to $\mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

9.2 Structure of Finite Dimensional Algebras

Chapter 10 Wedderburn-Artin-Theory

10.1 Simple,Semisimple Modules Modules

Definition 10.1.0.1

- A nonzero module M is **simple** or **irreducible** if it contains no proper non-zero submodule.
- A module M is called **semisimple** or **completely module** R-module if M is a direct sum of simple modules.

Definition 10.1.0.2 An R-module is **cyclic** with generator $m \in M$ such that

$$M = Rm$$

10.1.1 Simple Modules

Definition 10.1.1.1 An **irreducible representation** of an algebra A is defined as a representation

$$\varphi : A \to End_F(V)$$

which contains no proper nonzero subspace W of V that is invariant under $\varphi(a), a \in A$.

Equivalently, the associated module with structure $A \times V \to V, (a, v) \mapsto \rho(a)(v)$. The module V is irreducible.

♠ **Proposition 10.1.1.2** The following are equivalent for R-module M:

- M is simple;
- M is cyclic and every nonzero element is a generator;
- $M \cong R/I$ for some maximal left ideal I;

∘ **Proof**

- (1)⇒ (2) M is simple. Then for any $m \in M$, either $Rm = 0$ or $Rm = M$. Take

  $$N := \{m \in M \,|\, Rm = 0\}$$

  Then, N is a submodule, $N = 0$. This implies, that for every $m \neq 0$, $Rm = M$;
- (2)⇒ (3) Take

  $$\phi : R \to Rm \qquad r \mapsto rm$$

  And $I := Ker(\phi)$. I is a submodule of R, and $R/I \cong Rm = M$.
  I must be maixmal. If not,
- (3)⇒ (1) R/I contains no submodule.

□

♣ **Example 10.1.1.3** Let D be a division ring and V is vector space over D. Then D is simple$\Leftrightarrow dim_D(V) = 1$.

♣ **Example 10.1.1.4** Let V be a nonzero vector space over division ring D.

V can be regard as a module over $R = End_D(V)$, via $fv := f(v)$. We claim that V is a simple R-module: For each pair $v, w \in V$ with $v \neq 0$, there exist some f such that $f(v) = w$. This is true.

♣ **Example 10.1.1.5** Let R be a ring and L be an ideal. Then L is simple L-module iff L is minimal. Omitted.

Minimal Left Ideals

◇ **Lemma 10.1.1.6** Let R be a simple ring having a minimal left ideal L. Then every simple R-module is isomorphic to L.

∘ **Proof** ... □

Maximal Left Ideal

Definition 10.1.1.7 A left ideal U of a ring R is called a **maximal left ideal** if $U \neq R$ and U is not properly contained in a proper left ideal.

♣ **Example 10.1.1.8** Let $R = M_n(D)$, with D a division ring. And let

$$U = RE_{11} \oplus \cdots \oplus RE_{n-1,n-1}$$

It is easy to check: U is a maximal left ideal of R.
And:

$$R = U \oplus L$$

where $L = RE_{nn}$ is the minimal left ideal. Especially we have $R/U \oplus L$.

◇ **Lemma 10.1.1.9** Let R be a unital ring. Then every proper left ideal L of R is contained in a maximal left ideal.

∘ **Proof** Let $\mathfrak{S}$ be the set of all proper left ideals of R contains L,

$$S = \{I | I \text{ is left ideal and } L \subseteq I\}$$

Now, let $\{K_i | i \in I\}$ be a chain in S. Then

$$K = \cup K_i$$

is its upper bound. K is a left ideal, $1 \notin K \Rightarrow K \neq R$. By Zorn's lemma, □

◇ **Lemma 10.1.1.10** Let R be a ring. Every simple R-module is isomorphic to R/U for some maximal left ideal U of R.

Conversely, if U is a maximal ideal of R such that $R^2 \nsubseteq U$. Then R/U is simple R-module.

∘ **Proof** ... □

★ **Corollary 10.1.1.11** Every nonzero unital ring has a simple module.

∘ **Proof** ... □

10.1.2 Semisimple Modules

♣ **Example 10.1.2.1** ...

◇ **Lemma 10.1.2.2** If M is the sum of a family of simple submodules $\{M_i | i \in I\}$. Then, there exists a subset $K \subseteq I$ such that

$$M = \oplus_{k \in K} M_k$$

∘ **Proof** ... □

★ **Corollary 10.1.2.3** If a module is the sum of a family of simple submodules, then it is semisimple.

∘ **Proof** Omitted. □

◇ **Lemma 10.1.2.4** The following statements are equivalent for a nonzero module M:

- M is semisimple;
- Every submodule of M is a direct summand and $Rm \neq 0$ for every nonzero $m \in M$;

∘ **Proof** ... □

★ **Corollary 10.1.2.5** A **unital** module M is semisimple iff every submodule of M is a direct summand.

∘ **Proof** Omitted. □

Endomorphisms Rings of Semisimple Modules

▶ ***Theorem*** (Schur's Lemma) If M is a simple R-module. Then the endomorphism ring $End_R(M)$ is a division ring.

∘ **Proof** Let $f \in End_R(M)$ and $f \neq 0$. Then, $ker(f)$ is a submodule of M, $Ker(f) = 0$.
Then, $f \in Aut_R(M)$. f is inversable. There exists some $f^{-1} \in Aut_R(M)$. □

A slightly more general version:

- A nonzero homomorphism from an arbitrary module to a simple module is surjective,and;
- A nonzero homomorphism from a simple module to an arbitrary module is injective.

◇ **Lemma 10.1.2.6** If M, N are simple modules, when they are not isomorphic, there are no nonzero homomorphisms from one to another.

∘ **Proof** ... □

Semisimple Rings

10.1.3 Chain Conditions

10.2 Wedderburn-Artin Theorem